

Mathematics II

CSIT — 2nd Semester — 2079 — Answer Sheet
Subject Code: MTH168

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define system of linear equations. When a system of equations is consistent? Make echelon form to solve: $-2x_1 - 3x_2 + 4x_3 = 5$ $x_2 - 2x_3 = 4$ $x_1 + 3x_2 - x_3 = 2$ is consistent.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define linear transformation with an example. Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \end{bmatrix}$, $v = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and define a transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(x) = Ax$ then (a) find $T(v)$ (b) find $x \in \mathbb{R}^2$ whose image under T is b .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Find AB by block multiplication of the matrices $A = \begin{bmatrix} 2 & -3 & 1 & 0 & -4 \\ 1 & -5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 4 \\ 2 & -1 \\ -3 & 7 \\ 1 & 3 \\ 5 & -3 \end{bmatrix}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Find the least square solution of $Ax = c$, where $A = \begin{bmatrix} 1 & -3 & -3 \\ 1 & 5 & 1 \\ 1 & 7 & 2 \end{bmatrix}$, $c = \begin{bmatrix} 5 \\ -3 \\ 5 \end{bmatrix}$ and compute the associated least square error.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Determine the column of the matrix A are linearly independent, where $A = \begin{bmatrix} 3 & -3 & 6 \\ 0 & 2 & 4 \\ 0 & 3 & 0 \end{bmatrix}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Let $A = \begin{bmatrix} 1 & 5 \\ -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -5 \\ 3 & k \end{bmatrix}$. What value (s) of k, if any, will make $AB = BA$?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Evaluate the determinant of the matrix $\begin{bmatrix} 1 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. When two column vectors in $\mathbb{R}^2$ are equal? Given an example. Compute $(u + 3v, -u - 2v)$, where $u = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Prove that the two vectors u and v are perpendicular to each other if and only if the line through u is perpendicular bisector of the line segment from $-u$ to v .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Find the eigenvalue of $A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define null space of a matrix A . Let $A = \begin{bmatrix} 1 & -3 & 2 \\ -5 & 9 & -1 \end{bmatrix}$, and $v = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix}$, then show that v belongs to the null space matrix A .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Find the equation $y = a_0 + a_1 x$ of the least squares line that best fits the data points (2, 1), (5, 2), (7, 3), (8, 3)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.